

COMPUTATIONAL DYNAMICS - LECTURE 12

Recursive forward dynamics

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Recursive forward dynamics - Overview

- ▶ Recap
- ▶ Forward dynamics
- ▶ Generalized accelerations
- ▶ Decomposing the forward dynamics expression
- ▶ Body accelerations
- ▶ $O(N)$ recursive forward dynamics
- ▶ Alternative expressions
- ▶ Including gravity in the ATBI forward dynamics algorithm
- ▶ Including external forces in the ATBI forward dynamics algorithm
- ▶ Prescribed motion and hybrid dynamics
 - ▶ Passive and active system
- ▶ Hybrid dynamics solution
- ▶ Hybrid dynamics algorithm
- ▶ Summary



Recap

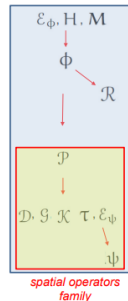


Recap

- ▶ Introduced spatial operators for Articulated body inertias
- ▶ Showed several operator identities
- ▶ Developed Innovations Factorization of the mass matrix
 - ▶ New properties for this includes: square and invertible factors
 - ▶ Can reduce the cost of forward dynamics from $O(N^3)$ to $O(N^2)$
- ▶ Developed expression for inverse of factors
- ▶ Developed operator expression for the mass matrix inverse

Spatial operators

- ▶ Velocity expression
 - ▶ Jacobian
 - ▶ Mass matrix Newton Euler factorization
 - ▶ Lyapunov equation for CRBs
 - ▶ Mass matrix decomposition
- ▶ Riccati equation for ATBI
 - ▶ Several operator identities
 - ▶ Mass matrix Innovations factorization
 - ▶ Mass matrix determinant
 - ▶ Mass matrix inverse and factorization



- ▶ Have started to build up a vocabulary of spatial operators that can be used to express and manipulate the structure of dynamics quantities
- ▶ Now we can see algebra part of SOA from the analytical transformations and simplifications possible using the operators.

Recursive Computational Algorithms

- ▶ Low-cost scatter/gather algorithms for examination of the spatial operator expressions has been developed
 - ▶ $O(N)$ Gather and scatter recursions pattern
 - ▶ $O(N)$ Body velocities scatter recursion
 - ▶ $O(N)$ CRBs gather recursion
 - ▶ $O(N^2)$ mass matrix computation
 - ▶ $O(N)$ Newton Euler scatter/gather inverse dynamics
 - ▶ $O(N^2)$ inverse dynamics based mass matrix
 - ▶ $O(N)$ CRBs based inverse dynamics
 - ▶ $O(N)$ ATBI gather recursion
 - ▶ $O(N^2)$ Forward dynamics

Forward Dynamics



System level equations of motion

- Operator level expressions

$$\mathcal{V} = \phi^T H^T \dot{\theta}$$

$$\alpha = \phi^T (H^T \ddot{\theta} + \mathbf{a})$$

$$\mathbf{f} = \phi (M \alpha + \mathbf{b})$$

$$\tau = H \mathbf{f}$$

- System/operator level equations of motion

$$\begin{aligned} \tau &= H \phi \left(M \phi^T (H^T \ddot{\theta} + \mathbf{a}) + \mathbf{b} \right) \\ &= M(\theta) \ddot{\theta} + \mathcal{C}(\theta, \dot{\theta}) \end{aligned}$$

Newton-Euler factorization

Mass matrix:

$$M(\theta) = \boxed{H \phi M \phi^T H^T} \in \mathbb{R}^{\mathcal{N} \times \mathcal{N}}$$

Coriolis terms:

$$\mathcal{C}(\theta, \dot{\theta}) \triangleq H \phi (M \phi^T \mathbf{a} + \mathbf{b}) \in \mathbb{R}^{\mathcal{N}}$$



Inverse and Forward dynamics

- ▶ Inverse dynamics:

- ▶ Given the state and generalized accelerations the equations of motion are used to compute the generalized forces

$$\tau = M(\theta) \ddot{\theta} + \mathcal{C}(\theta, \dot{\theta})$$

- ▶ Important for control applications

- ▶ Forward dynamics:

- ▶ Given the state and generalized forces the equations of motion are used to compute the generalized accelerations

$$\ddot{\theta} = M^{-1}(\tau - \mathcal{C})$$

- ▶ Important for simulation applications

$O(N^3)$ forward dynamics

- ▶ Forward dynamics involves computing the generalized accelerations

$$\ddot{\theta} = M^{-1} (\tau - \mathcal{C})$$

- ▶ The conventional approach would be to determine the mass matrix and then solve the above matrix equation
- ▶ The conventional approach is of $O(N^3)$ computational complexity
- ▶ Can we do better?

Spatial operator identities recap

- ▶ For improving the forward dynamics computational complexity the operator identities seen here are very useful for transforming and simplifying operator expressions
- ▶ We will see their use in later derivations

$$\psi^{-1} - \phi^{-1} = \mathcal{K} H$$

$$\psi^{-1} \phi = I + \mathcal{K} H \phi$$

$$\phi \psi^{-1} = I + \phi \mathcal{K} H$$

$$\phi^{-1} \psi = I - \mathcal{K} H \psi$$

$$\psi \phi^{-1} = I - \psi \mathcal{K} H$$

$$(I - H \psi \mathcal{K}) H \phi = H \psi$$

$$\phi \mathcal{K} (I - H \psi \mathcal{K}) = \psi \mathcal{K}$$

$$(I + H \phi \mathcal{K}) H \psi = H \phi$$

$$\psi \mathcal{K} (I + H \phi \mathcal{K}) = \phi \mathcal{K}$$

Other operator identity

- ▶ Earlier mass matrix expression (Newton Euler factorisation)

Dense

$$M(\theta) \stackrel{\Delta}{=} H \phi M \phi^T H^T$$

- ▶ Versus similar expression

Block-diagonal

$$\mathcal{D} = H \psi M \psi^T H^T$$

- ▶ Complex product of spatial operators collapses into just $\mathcal{D}$!
- ▶ The only difference is the use of ψ instead of ϕ !

$O(N^2)$ forward dynamics using Innovations Factorization - Recall

$$M = (I + H \phi K) \mathcal{D} (I + H \phi K)^T$$

- ▶ Applying the previous identities the expression for the Innovations Factorization is found.
- ▶ All the factors are square in the Innovations Factorization
 - ▶ $(I + H \phi K)$ Lower triangular with identity along the block-diagonal
 - ▶ $\mathcal{D}$ Block-diagonal
 - ▶ $(I + H \phi K)^T$ Upper triangular with identity along the block-diagonal
- ▶ So want to look into inverting the mass matrix by inverting its factors
- ▶ Since $\mathcal{D}$ is block-diagonal it is easy to invert
- ▶ We will thus focus on inverting $(I + H \phi K)$

$O(N^2)$ forward dynamics using Innovations Factorization - Recall

$$M = (I + H \phi K) \mathcal{D} (I + H \phi K)^T$$

- ▶ Forward dynamics involves computing the generalized accelerations

$$\ddot{\theta} = M^{-1} (\tau - C)$$

- ▶ With the Newton Euler factorization the cost was $O(N^3)$
 - ▶ The Innovations Factorization can be computed at $O(N^2)$ cost
- ▶ The Innovations Factorization can compute $\ddot{\theta} = M^{-1} (\tau - C)$ at $O(N^2)$ cost as well
- ▶ We have reduced the cost from $O(N^3)$ to $O(N^2)$.
- ▶ This is progress - but we can do better.



Mass Matrix Factorization and Inversion

- ▶ Analytical Newton-Euler factorization of the mass matrix

$$M = H \phi M \phi^T H^T$$

- ▶ Analytical Innovations factorization of the mass matrix

$$M = (I + H \phi \mathcal{K}) \mathcal{D} (I + H \phi \mathcal{K})^T$$

- ▶ Analytical operator expression for the mass matrix inverse

$$\begin{aligned} M^{-1} &= (I - H \psi \mathcal{K})^T \mathcal{D}^{-1} (I - H \psi \mathcal{K}) \\ (I + H \phi \mathcal{K})^{-1} &= (I - H \psi \mathcal{K}) \end{aligned}$$

- ▶ This is where we ended last, we are going to use this to simplify the forward dynamics

Forward dynamics expression

- ▶ Solving for the generalized accelerations

$$\ddot{\theta} = M^{-1}(\tau - \mathcal{C})$$

- ▶ Operator expression for coriolis term and the mass matrix inverse

$$\mathcal{C}(\theta, \dot{\theta}) \triangleq H \phi (M \phi^T \mathbf{a} + \mathbf{b})$$

$$M^{-1} = (I - H \psi \mathcal{K})^T \mathcal{D}^{-1} (I - H \psi \mathcal{K})$$

- ▶ Collected expression for generalized accelerations

$$\begin{aligned}\ddot{\theta} &= (I - H \psi \mathcal{K})^T \mathcal{D}^{-1} (I - H \psi \mathcal{K}) (\tau - \mathcal{C}) \\ &= (I - H \psi \mathcal{K})^T \mathcal{D}^{-1} (I - H \psi \mathcal{K}) (\tau - H \phi (M \phi^T \mathbf{a} + \mathbf{b}))\end{aligned}$$

Forward dynamics expression

- Solving for the generalized accelerations

$$\begin{aligned}\ddot{\theta} &= (I - H \psi \mathcal{K})^T \mathcal{D}^{-1} (I - H \psi \mathcal{K}) (\tau - \mathcal{C}) \\ &= (I - H \psi \mathcal{K})^T \mathcal{D}^{-1} (I - H \psi \mathcal{K}) (\tau - H \phi (M \phi^T \mathbf{a} + \mathbf{b}))\end{aligned}$$

- Have to do four sweeps to gather/scatter sweeps to compute $\ddot{\theta}$
- See if we can simplify this to reduce the cost
- This is complex so there is still work to do!
- Focus on simplifying the last part of this expression

Generalized Accelerations



Simplified expression for $\ddot{\theta}$

$$\begin{aligned}\ddot{\theta} &= (I - H \psi \mathcal{K})^T \mathcal{D}^{-1} (I - H \psi \mathcal{K}) (\tau - \mathcal{C}) \\ &= (I - H \psi \mathcal{K})^T \mathcal{D}^{-1} (I - H \psi \mathcal{K}) \left(\tau - \boxed{H \phi (M \phi^T \mathbf{a} + \mathbf{b})} \right)\end{aligned}$$

Coriolis terms

- Claim is that there is a simplification of the above expression

$$\ddot{\theta} = (I - H \psi \mathcal{K})^T \mathcal{D}^{-1} (\tau - H \psi (K \tau + \boxed{P \mathbf{a} + \mathbf{b}})) - K^T \psi^T \mathbf{a}$$

Coriolis terms

- This expression is much simpler
- The products of spatial operators has been reduced from 4 to 2!
- Now only has to do two sweeps!



Derivation of $\ddot{\theta}$ expression

► Step 1:

$$\begin{aligned}\ddot{\theta} &= (I - H \psi \mathcal{K})^T \mathcal{D}^{-1} (I - H \psi \mathcal{K}) (\tau - \mathcal{C}) \\ &= (I - H \psi \mathcal{K})^T \mathcal{D}^{-1} (I - H \psi \mathcal{K}) (\tau - H \phi (M \phi^T \mathbf{a} + \mathbf{b})) \\ &= (I - H \psi \mathcal{K})^T \mathcal{D}^{-1} (I - H \psi \mathcal{K}) \tau \\ &\quad - (I - H \psi \mathcal{K})^T \mathcal{D}^{-1} \boxed{(I - H \psi \mathcal{K}) H \phi (M \phi^T \mathbf{a} + \mathbf{b})}\end{aligned}$$

Involves multiple operator products. Focus on this term!

$$\mathcal{C}(\theta, \dot{\theta}) \triangleq H \phi (M \phi^T \mathbf{a} + \mathbf{b})$$



Derivation of $\ddot{\theta}$ expression (contd)

► Step 2:

$$\begin{aligned}(I - H \psi \mathcal{K}) H \phi (M \phi^T \mathbf{a} + \mathbf{b}) &= H \psi (M \phi^T \mathbf{a} + \mathbf{b}) \\&= H \left((\psi P + P \tilde{\phi}^T) \mathbf{a} + \psi \mathbf{b} \right) \\&= (H \psi P + \mathcal{D} \mathcal{K}^T \phi^T) \mathbf{a} + H \psi \mathbf{b} \\&= H \psi (P \mathbf{a} + \mathbf{b}) + \mathcal{D} \mathcal{K}^T \phi^T \mathbf{a}\end{aligned}$$

$$(I - H \psi \mathcal{K}) H \phi = H \psi \qquad \phi M \psi^T = P + \tilde{\phi} P + P \tilde{\psi}^T$$

► Much simpler – The products spatial operators has been reduced from 3 to solitary occurrences!



Derivation of $\ddot{\theta}$ expression (contd)

► Step 3:

$$\begin{aligned}\ddot{\theta} &= (I - H \psi \mathcal{K})^T \mathcal{D}^{-1} (I - H \psi \mathcal{K}) H \phi (M \phi^T \mathbf{a} + \mathbf{b}) \\ &= (I - H \psi \mathcal{K})^T \mathcal{D}^{-1} (H \psi (P \mathbf{a} + \mathbf{b}) + \mathcal{D} \mathcal{K}^T \phi^T \mathbf{a}) \\ &= (I - H \psi \mathcal{K})^T \mathcal{D}^{-1} H \psi (P \mathbf{a} + \mathbf{b}) + \mathcal{K}^T \phi^T \mathbf{a}\end{aligned}$$

$$\phi \mathcal{K} (I - H \psi \mathcal{K}) = \psi \mathcal{K}$$

Simplified expression for $\ddot{\theta}$

- ▶ The original expression for the generalized accelerations

$$\begin{aligned}\ddot{\theta} = & (I - H \psi \mathcal{K})^T \mathcal{D}^{-1} (I - H \psi \mathcal{K}) \tau \\ & - (I - H \psi \mathcal{K})^T \mathcal{D}^{-1} (I - H \psi \mathcal{K}) H \phi (M \phi^T \mathbf{a} + \mathbf{b})\end{aligned}$$

- ▶ Operator manipulations helped simplify the generalized accelerations expression

$$\ddot{\theta} = (I - H \psi \mathcal{K})^T \mathcal{D}^{-1} (\tau - H \psi (\mathcal{K} \tau + P \mathbf{a} + \mathbf{b})) - \mathcal{K}^T \psi^T \mathbf{a}$$

- ▶ Reduced the number of multiple operator products from 4 to 2.
- ▶ This is much simpler!



Decomposing the Forward Dynamics Expression



Decomposing the $\ddot{\theta}$ expression

► Breaking down the expression:

$$\ddot{\theta} = (I - H \psi \mathcal{K})^T \left[\mathcal{D}^{-1} \left(\tau - H \underbrace{\psi (\mathcal{K} \tau + P \mathbf{a} + \mathbf{b})}_{\xi} \right) \right]_{\epsilon} - \mathcal{K}^T \psi^T \mathbf{a}$$

ν

$$\xi \triangleq \psi (\mathcal{K} \tau + P \mathbf{a} + \mathbf{b})$$

$$\epsilon \triangleq \tau - H \xi = \tau - H \psi (\mathcal{K} \tau + P \mathbf{a} + \mathbf{b})$$

$$\nu \triangleq \mathcal{D}^{-1} \epsilon = \mathcal{D}^{-1} (\tau - H \psi (\mathcal{K} \tau + P \mathbf{a} + \mathbf{b}))$$

$$\ddot{\theta} = (I - H \psi \mathcal{K})^T \nu - \mathcal{K}^T \psi^T \mathbf{a}$$

Alternative expression for ξ

► Claim:

$$\xi \triangleq \psi(\mathcal{K} \tau + P \mathbf{a} + \mathbf{b})$$

$$\boxed{\xi = \phi(\mathcal{K} \epsilon + P \mathbf{a} + \mathbf{b})}$$

Maps into an $O(N)$ recursive gather algorithm

► Derivation:

$$\epsilon = (I - H \psi \mathcal{K}) \tau - H \psi(P \mathbf{a} + \mathbf{b})$$

$$\begin{aligned} \tau &= (I - H \psi \mathcal{K})^{-1} (\epsilon + H \psi(P \mathbf{a} + \mathbf{b})) \\ &= (I + H \phi \mathcal{K}) \epsilon + \epsilon \phi(P \mathbf{a} + \mathbf{b}) \end{aligned}$$



Alternative expression for ξ (contd)

► Claim:

$$\xi \stackrel{\Delta}{=} \psi(\mathcal{K} \tau + P \mathfrak{a} + \mathfrak{b})$$

► Derivation:

$$\xi = \phi(\mathcal{K} \epsilon + P \mathfrak{a} + \mathfrak{b})$$

$$\begin{aligned}\xi &= \psi(\mathcal{K} \tau + P \mathfrak{a} + \mathfrak{b}) \\ &= \psi \mathcal{K} (I + H \phi \mathcal{K}) \epsilon + (\psi \mathcal{K} H \phi + \psi) (P \mathfrak{a} + \mathfrak{b}) \\ &= \phi \mathcal{K} \epsilon + (\phi - \psi + \psi) (P \mathfrak{a} + \mathfrak{b}) \\ &= \phi(\mathcal{K} \epsilon + P \mathfrak{a} + \mathfrak{b})\end{aligned}$$

$$\tau = (I + H \phi \mathcal{K}) \epsilon + \epsilon \phi (P \mathfrak{a} + \mathfrak{b})$$

Defining ξ^+ and re-expressing ξ

► Have

$$\xi = \phi (\mathcal{K} \epsilon + P \mathbf{a} + \mathbf{b}) \qquad \xi^+ \triangleq \xi + \mathcal{G} \epsilon$$

► Claim:

$$\xi = \epsilon_\phi \xi^+ + P \mathbf{a} + \mathbf{b}$$

► Derivation:

$$\begin{aligned} \phi^{-1} &= (I - \epsilon_\phi) && \text{Pre multiplying and using this} \\ \xi &= \epsilon_\phi \xi + (\mathcal{K} \epsilon + P \mathbf{a} + \mathbf{b}) = \epsilon_\phi (\xi + \mathcal{G} \epsilon) + P \mathbf{a} + \mathbf{b} \\ &= \epsilon_\phi \xi^+ + P \mathbf{a} + \mathbf{b} \end{aligned}$$

Alternative expression for ξ^+

▮ Have

$$\xi^+ \triangleq \xi + \mathcal{G} \epsilon$$

▮ Claim:

$$\xi^+ \triangleq \phi (\mathcal{G} \epsilon + P \mathbf{a} + \mathbf{b})$$

▮ Derivation:

$$\begin{aligned} \xi^+ &= \xi + \mathcal{G} \epsilon = \phi(\mathcal{K} \epsilon + P \mathbf{a} + \mathbf{b}) + \mathcal{G} \epsilon \\ &= \tilde{\phi} \mathcal{G} \epsilon + \phi (P \mathbf{a} + \mathbf{b}) + \mathcal{G} \epsilon \\ &= \phi (\mathcal{G} \epsilon + P \mathbf{a} + \mathbf{b}) \end{aligned}$$

Body accelerations



Body acceleration expression

- ▶ Mapping from generalised accelerations to body accelerations

$$\alpha = \phi^T \left(H^T \ddot{\theta} + \mathbf{a} \right)$$

- ▶ Claim:

$$\alpha = \psi^T \left(H^T \nu + \mathbf{a} \right)$$

$O(N)$ recursive scatter algorithm

- ▶ ν is a function of generalised forces not generalised accelerations

Body acceleration expression (contd)

$$\alpha = \phi^T \left(H^T \ddot{\theta} + \mathbf{a} \right)$$

► Claim:

$$\alpha = \psi^T \left(H^T \nu + \mathbf{a} \right)$$

► Derivation:

$$\begin{aligned} \alpha &= \phi^T \left(H^T \ddot{\theta} + \mathbf{a} \right) = \phi^T H^T \left((I - H \psi \mathcal{K})^T \nu - \mathcal{K}^T \psi^T \mathbf{a} \right) + \phi^T \mathbf{a} \\ &= \psi^T H^T \nu - \phi^T H^T \mathcal{K}^T \psi^T \mathbf{a} + \psi^T \mathbf{a} = \psi^T H^T \nu + \psi^T \mathbf{a} = \psi^T \left(H^T \nu + \mathbf{a} \right) \end{aligned}$$

$$\ddot{\theta} = (I - H \psi \mathcal{K})^T \nu - \mathcal{K}^T \psi^T \mathbf{a} \qquad (I - H \psi \mathcal{K}) H \phi = H \psi$$

$$\psi^{-1} - \phi^{-1} = \mathcal{K} H$$

Generalized acceleration expression

$$\alpha^+ \triangleq \text{col}(\alpha^+(k))_{k=1}^n = \epsilon_\phi^T \alpha \qquad \ddot{\theta} = (I - H \psi \mathcal{K})^T \nu - \mathcal{K}^T \psi^T \mathbf{a}$$

► Claim:

$$\ddot{\theta} = \nu - \mathcal{K}^T \alpha = \nu - \mathcal{G}^T \alpha^+$$

► Derivation:

$$\begin{aligned} \ddot{\theta} &= (I - H \psi \mathcal{K})^T \nu - \mathcal{K}^T \psi^T \mathbf{a} \\ &= \nu - \mathcal{K}^T \psi^T H^T \nu - \mathcal{K}^T \psi^T \mathbf{a} \\ &= \nu - \mathcal{K}^T \alpha \end{aligned}$$

$O(N)$ Recursive Forward Dynamics

Overall decomposed expressions

- ▶ Collecting the expressions we have derived so far
- ▶ We are going to map these expressions into recursive algorithms

$$\xi = \epsilon_{\phi} \xi^+ + P \mathbf{a} + \mathbf{b}$$

$$\xi^+ = \xi + \mathcal{G} \epsilon$$

$$\epsilon = \tau - H \xi$$

$$\nu = \mathcal{D}^{-1} \epsilon$$

$$\alpha^+ = \epsilon_{\phi}^T \alpha$$

$$\ddot{\theta} = \nu - \mathcal{G}^T \alpha^+$$

$$\alpha = \alpha^+ + H^T \ddot{\theta} + \mathbf{a}$$

$O(N)$ ATBI forward dynamics algorithm

- Gather sweep
- First part of the gather sweep is known from the ATBI recursion

$$P^+(0) = 0, \quad \xi^+(0) = 0, \quad \tau(0) = 0, \quad \bar{\tau}(0) = 0$$

for $k = 1 \dots n$

$$P(k) = \phi(k, k-1) P^+(k-1) \phi^T(k, k-1) + M(k)$$
$$D(k) = H(k) P(k) H^T(k)$$
$$\mathcal{G}(k) = P(k) H^T(k) \mathcal{D}^{-1}(k)$$
$$\bar{\tau}(k) = I - \mathcal{G}(k) H(k)$$
$$P^+(k) = \bar{\tau}(k) P(k)$$
$$\xi(k) = \phi(k, k-1) \xi^+(k-1) + P(k) \mathbf{a}(k) + \mathbf{b}(k)$$
$$\epsilon(k) = \tau(k) - H(k) \xi(k)$$
$$\nu(k) = \mathcal{D}^{-1}(k) \epsilon(k)$$
$$\xi^+ = \xi(k) + \mathcal{G}(k) \epsilon(k)$$

end loop

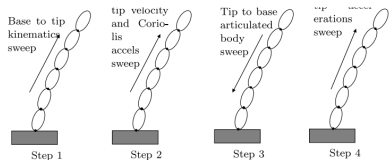
$O(N)$ ATBI forward dynamics algorithm

- ▶ Scatter sweep
- ▶ $\ddot{\theta}$ is of $O(N)$ computational complexity
- ▶ This is the fastest available algorithm

```
α(n + 1) = 0
for k = n . . . 1
    α+(k) = φT(k + 1, k) α(k + 1)
     $\ddot{\theta}(k) = \nu(k) - \mathcal{G}^T(k) \alpha^+(k)$ 
    α(k) = α+(k) + HT(k)  $\ddot{\theta}(k)$  + a(k)
end loop
```

Structure of the ATBI $O(N)$ recursions

- ▶ Step 1: Velocities and poses of the bodies (kinematics)
- ▶ Step 2: Compute the Coriolis terms $\mathbf{a}$ and $\mathbf{b}$
- ▶ Step 3: Gather from previous slides
- ▶ Step 4: Scatter from previous slides
- ▶ The first two steps can be combined in one recursion
- ▶ All steps are of $O(N)$ cost
- ▶ Structure of the $O(N)$ AB Algorithm 7.1 for serial-chain forward dynamics

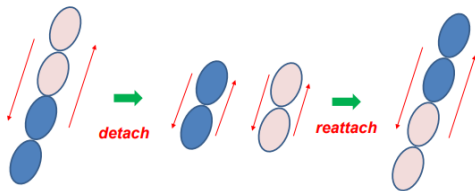


Comments on the ATBI algorithm

- ▶ The ATBI algorithm is of $O(N)$ cost and therefore scales linearly with the number of degrees of freedom
- ▶ It is the fastest available to date
- ▶ No explicit computation of the operators is required
- ▶ It is not required to compute the mass matrix inverse or the mass matrix themselves
- ▶ The algorithm applies to any size system
- ▶ The algorithm consists of gather and scatter sweeps, and these easily accommodate changes to the system from the addition or removal of bodies
- ▶ The ATBI algorithm continues to work for more general branched systems (later)



Handling structural configuration changes



- ▶ Multibody methods often have to work hard to handle structural changes – when they can
- ▶ The scatter and gather recursions however follow the instantaneous topological configuration (even addition and removal of bodies)
- ▶ Hence the SOA structure-based algorithms accommodate configuration changes automatically without missing a heartbeat!

Alternative expressions



Alternative expressions for body accelerations

- Claim:

$$\alpha = \bar{\tau}^T \alpha^+ + H^T \nu + \mathbf{a} \qquad \nu = \mathcal{G}^T (\alpha - \mathbf{a})$$

- Derivation

$$\begin{aligned} \alpha &= \psi^T (H^T \nu + \mathbf{a}) = \tilde{\psi}^T (H^T \nu + \mathbf{a}) + (H^T \nu + \mathbf{a}) \\ &= \epsilon_{\tilde{\psi}}^T \psi^T (H^T \nu + \mathbf{a}) + (H^T \nu + \mathbf{a}) = \epsilon_{\tilde{\psi}}^T \alpha + H^T \nu + \mathbf{a} \\ &= \bar{\tau}^T \alpha^+ + H^T \nu + \mathbf{a} \end{aligned}$$

- The second equation is obtained by multiplying the first by $\mathcal{G}^T$

Inter-body forces

- ▶ In the absolute coordinate methods, the inter-body forces are the Lagrange multipliers associated with the constraints.
- ▶ These are not computed by the $O(N)$ algorithm, but can be readily computed if needed

Inter-body forces

- Claim:

$$\mathbf{f} = P (\alpha - \mathbf{a}) + \xi = P^+ (\alpha^+ + \xi^+)$$

- Derivation

$$\begin{aligned}\mathbf{f} &= \phi (M \alpha + \mathbf{b}) = \phi (M \psi^T (H^T \nu + \mathbf{a}) + \mathbf{b}) \\ &= \left(\tilde{\phi} P + P \psi \right) (H^T \nu + \mathbf{a}) + \phi \mathbf{b} = \tilde{\phi} P H^T \nu + \tilde{\phi} P \mathbf{a} + P \alpha + \phi \mathbf{b} \\ &= \phi \mathcal{K} \epsilon + \phi P \mathbf{a} - P \mathbf{a} + P \alpha + \phi \mathbf{b} = P (\alpha - \mathbf{a}) + \xi\end{aligned}$$

$$\xi = \phi (\mathcal{K} \epsilon + P \mathbf{a} + \mathbf{b})$$

- Similarly can show second equality.



Estimation theory parallels

- Gather sweep
- ξ : casual filtering sweep

$$P^+(0) = 0, \quad \xi^+(0) = 0, \quad \tau(0) = 0, \quad \bar{\tau}(0) = 0$$

for $k = 1 \dots n$

$$P(k) = \phi(k, k-1) P^+(k-1) \phi^T(k, k-1) + M(k)$$

$$D(k) = H(k) P(k) H^T(k)$$

$$\mathcal{G}(k) = P(k) H^T(k) \mathcal{D}^{-1}(k)$$

$$\bar{\tau}(k) = I - \mathcal{G}(k) H(k)$$

$$P^+(k) = \bar{\tau}(k) P(k)$$

$$\boxed{\xi(k)} = \phi(k, k-1) \xi^+(k-1) + P(k) \mathbf{a}(k) + \mathbf{b}(k)$$

Optimal filter estimate

$$\epsilon(k) = \tau(k) - H(k) \xi(k)$$

$$\nu(k) = \mathcal{D}^{-1}(k) \epsilon(k)$$

$$\xi^+ = \xi(k) + \mathcal{G}(k) \epsilon(k)$$

end loop



Estimation theory parallels

- Scatter sweep
- $\ddot{\theta}$: anti-causal co-state sweep
- ξ : casual filtering sweep

```

 $\alpha(n+1) = 0$ 
for  $k = n \dots 1$ 
     $\alpha^+(k) = \phi^T(k+1, k) \alpha(k+1)$ 
     $\ddot{\theta}(k) = \nu(k) - \mathcal{G}^T(k) \alpha^+(k)$ 
     $\boxed{\alpha(k)} = \alpha^+(k) + H^T(k) \ddot{\theta}(k) + \mathbf{a}(k)$ 
    co-state
end loop
    
```

$$\boxed{\mathbf{f}} = P(\alpha + \mathbf{a}) + \xi = P^+ \alpha^+ + \xi^+$$

Optimal smoothed estimate



Including Gravity in the ATBI Forward Dynamics Algorithm



Including gravitational acceleration

- ▶ The Coriolis term with gravity term is

$$\mathcal{C}(\theta, \dot{\theta}) = H \phi \left(M \phi^T (-I + E^T \mathbf{g}) + \mathbf{b} \right)$$

- ▶ Claim:

$$\ddot{\theta} = (I - H \psi \mathcal{K})^T \left(D^{-1}(\tau - H \psi (\mathcal{K} \tau + P \mathbf{a} + \mathbf{b})) - \mathcal{G}^T \bar{E}^T \mathbf{g} \right) - \mathcal{K}^T \psi^T \mathbf{a}$$

- ▶ Derivation:
- ▶ Define $\mathbf{a}' \triangleq \mathbf{a} + E^T \mathbf{g}$ and use in place of $\mathbf{a}$ in the original proof of the forward dynamics expression and do further simplifications.

ATBI algorithm update for gravity

- Define

$$\bar{\nu} \triangleq \nu - \mathcal{G}^T \bar{E}^T \mathfrak{g}$$

- Claim:

$$\alpha = \psi^T (H^T \bar{\nu} + \mathfrak{a}) \qquad \ddot{\theta} = (I - H \psi \mathcal{K})^T \bar{\nu} - \mathcal{K}^T \psi^T \mathfrak{a} = \bar{\nu} - \mathcal{G}^T \alpha^+$$

- Derivation:

$$\begin{aligned} \alpha &= \phi^T (H^T \ddot{\theta} + \mathfrak{a}) = \psi^T (H^T \nu + \mathfrak{a}) - \phi^T H^T (I - H \psi \mathcal{K})^T \mathcal{G}^T \bar{E}^T \mathfrak{g} \\ &= \psi^T (H^T \nu + \mathfrak{a}) - \psi^T H^T \mathcal{G}^T \bar{E}^T \mathfrak{g} = \psi^T (H^T (\nu - \mathcal{G}^T \bar{E}^T \mathfrak{g}) + \mathfrak{a}) \\ &= \psi^T (H^T \bar{\nu} + \mathfrak{a}) \end{aligned}$$



- Replace ν with $\bar{\nu}$ in the original $O(N)$ algorithm

Updated inter-body force computation with gravity

- Earlier: $\mathbf{f} = P (\alpha - \mathbf{a}) + \xi = P^+ (\alpha^+ + \xi^+)$
- ▶ Claim: $\mathbf{f} = P (\alpha + \bar{E}^T \mathbf{g} - \mathbf{a}) + \xi = P^+ (\alpha^+ + \bar{E}^T \mathbf{g} + \xi^+)$
- ▶ Derivation

$$\begin{aligned}
 \mathbf{f} &= \phi \left(M (\alpha + \bar{E}^T \mathbf{g}) + \mathbf{b} \right) = \phi \left(M (\psi^T (H^T \nu + \mathbf{a}) + \bar{E}^T \mathbf{g}) + \mathbf{b} \right) \\
 &= (\tilde{\phi} P + P \psi) (H^T \nu + \mathbf{a}) + \phi \mathbf{b} + \phi M \bar{E}^T \mathbf{g} \\
 &= \tilde{\phi} P H^T \nu + \tilde{\phi} P \mathbf{a} + P \alpha + \phi \mathbf{b} + (\phi M - \tilde{\phi} P H^T \mathcal{G}^T) \bar{E}^T \mathbf{g} \\
 &= \phi \mathcal{K} \epsilon + \phi P \mathbf{a} - P \mathbf{a} + P \alpha + \phi \mathbf{b} + (\phi M \phi^T - \tilde{\phi} P H^T \mathcal{G}^T \phi^T) E^T \mathbf{g} \\
 &= P (\alpha - \mathbf{a}) + \xi + (\tilde{\phi} P + P \phi^T - \tilde{\phi} P H^T \mathcal{G}^T) E^T \mathbf{g} \\
 &= P (\alpha + \bar{E}^T \mathbf{g} - \mathbf{a}) + \xi + (\tilde{\phi} P - \tilde{\phi} P H^T \mathcal{G}^T) E^T \mathbf{g} \\
 &= P (\alpha + \bar{E}^T \mathbf{g} - \mathbf{a}) + \xi + \tilde{\phi} P \bar{\tau}^T E^T \mathbf{g} \\
 &= P (\alpha + \bar{E}^T \mathbf{g} - \mathbf{a}) + \xi
 \end{aligned}$$

Including external forces in the ATBI Forward Dynamics Algorithm



Including external forces

- The effect of external forces at nodes on the individual bodies can be accommodated by using the following modified step in the recursive algorithm:

$$\xi(k) = \phi(k, k-1) \xi^+(k-1) + P(k) \alpha(k) + \mathfrak{b}(k) - \boxed{\sum_i \phi(B_k, O_k^i) \mathfrak{f}_{ext}(k)}$$

External forces contributing for the kth body

Overall ATBI algorithm with gravity and external forces

➤ Gather sweep

$$P^+(0) = 0, \quad \xi^+(0) = 0, \quad \tau(0) = 0, \quad \bar{\tau}(0) = 0$$

for $k = 1 \dots n$

$$P(k) = \phi(k, k-1) P^+(k-1) \phi^T(k, k-1) + M(k)$$

$$D(k) = H(k) P(k) H^T(k)$$

$$\mathcal{G}(k) = P(k) H^T(k) \mathcal{D}^{-1}(k)$$

$$\bar{\tau}(k) = I - \mathcal{G}(k) H(k)$$

$$P^+(k) = \bar{\tau}(k) P(k)$$

$$\xi(k) = \phi(k, k-1) \xi^+(k-1) + P(k) \mathbf{a}(k) + \mathbf{b}(k) - \underbrace{\sum_i \phi(B_k, O_k^i) \mathbf{f}_{ext}(k)}_{\text{External forces}}$$

$$\epsilon(k) = \tau(k) - H(k) \xi(k)$$

$$\nu(k) = \mathcal{D}^{-1}(k) \epsilon(k)$$

$$\xi^+ = \xi(k) + \mathcal{G}(k) \epsilon(k)$$

end loop

Overall ATBI algorithm with gravity and external forces

► Scatter sweep

```
 $\alpha(n+1) = 0$   
for  $k = n \dots 1$   
   $\alpha^+(k) = \phi^T(k+1, k) \alpha(k+1)$   
   $\bar{\nu}(k) = \nu(k) - \mathcal{G}^T(k) \mathbf{g}$   
   $\ddot{\theta}(k) = \bar{\nu}(k) - \mathcal{G}^T(k) \alpha^+(k)$   
   $\alpha(k) = \alpha^+(k) + H^T(k) \ddot{\theta}(k) + \mathbf{a}(k)$   
end loop
```

Progression of mass matrix expressions and related algorithms

- The progressive derivation of spatial operator expressions leads to family of algorithms and reduction in computational cost.

$$\mathcal{M} = H\phi M\phi^* H^* \rightarrow O(N) \text{ Newton-Euler Inverse Dynamics}$$

$$O(N^2) \text{ Composite Body Algorithm for } \mathcal{M}$$

$$O(N^3) \text{ Forward Dynamics}$$

$$\mathcal{M} = [I + H\phi K]D[I + H\phi K]^* \rightarrow O(N^2) \text{ Forward Dynamics}$$

$$[I + H\phi K]^{-1} = [I - H\psi K]$$

$$\mathcal{M}^{-1} = [I - H\psi K]^* D^{-1} [I - H\psi K] \rightarrow O(N) \text{ Articulated Body Forward Dynamics}$$

$$O(N^2) \text{ Computation of } \mathcal{M}^{-1}$$

Minimal coordinates and the ATBI algorithm

- ▶ The minimal coordinates approach is the remarkable existence of the $O(N)$ ATBI algorithm – in contrast with the $O(N^3)$ cost of conventional methods
 - ▶ The ATBI algorithm can be derived directly – without the use of SOA operators.
 - ▶ First versions by Vereshchagin and Armstrong in 1970's, and more general versions by Featherstone and Rodriguez in 1980s
 - ▶ Several others were published soon after - all shown to be essentially the same ATBI algorithm with different notation
- ▶ There is a lot more to minimal coordinate dynamics than just the ATBI algorithm

Minimal coordinates and the ATBI algorithm (contd)

- ▶ Exploring the broader analytical **structure** of minimal coordinate dynamics, and developing a unifying approach to low-cost algorithms (including ATBI algorithm) is the goal of the broader SOA approach
- ▶ Despite its computational benefits, the ATBI algorithm is not used as widely as it should be
- ▶ Without proper mathematical tools, generalizing the ATBI algorithms from scratch to apply to the broad class of multibody systems can be very daunting
 - ▶ Body flexibility
 - ▶ General branching
 - ▶ Closed chain and prescribed motion
 - ▶ Configuration changes

Minimal coordinates and the ATBI algorithm (contd)

- ▶ The use of the ATBI algorithm has been confined by and large to rigid body serial chains for robotic arm applications
- ▶ SOA provides the mathematical machinery to address this issue, so the work is done once to learn the framework, and then generalizations and algorithms become easy

Prescribed Motion and Hybrid Dynamics



Inverse and Forward dynamics

- ▶ Inverse dynamics:

- ▶ Given the state, and generalized accelerations, use the equations of motion to compute the generalized forces

$$\tau = M(\theta) \ddot{\theta} + \mathcal{C}(\theta, \dot{\theta})$$

- ▶ N general accelerations known, N generalised forces unknown

- ▶ Forward dynamics:

- ▶ Given the state, and generalized forces, solve the equations of motion to compute the generalized accelerations

$$\ddot{\theta} = M^{-1}(\tau - \mathcal{C})$$

- ▶ N generalised forces known, N generalised accelerations unknown

Hybrid dynamics

$$\tau = M(\theta) \ddot{\theta} + \mathcal{C}(\theta, \dot{\theta})$$

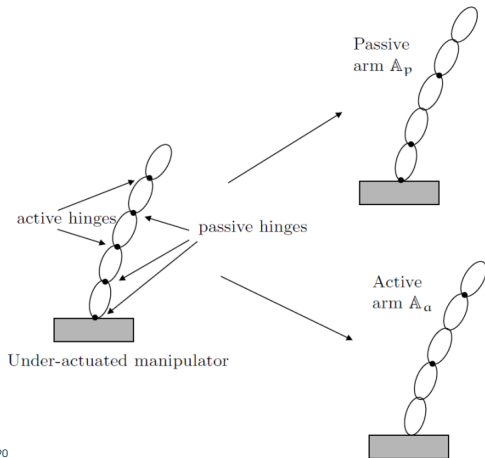
- ▶ Assume mix of N generalized accelerations and generalized forces are known and the complementary N generalized accelerations and forces are to be determined
- ▶ When the known are all generalised accelerations we have inverse dynamics, and when the known are all generalized forces we have forward dynamics
- ▶ Hybrid dynamics is a generalization that covers both forward and inverse dynamics
- ▶ The known generalized accelerations are also referred to by 'prescribed motion' in the context of forward dynamics

Examples of Hybrid dynamics applications

- ▶ Control of under-actuated systems
 - ▶ Fewer actuators than DoFs
 - ▶ Most mobile ground robots are under-actuated such as vehicles and walking humanoids as they tend to have all their hinges actuated but their chassis tend to have 6 DoF that is not actuated
- ▶ Docking, separation, jettison configuration change scenarios, eg.
 - ▶ Heat-shield separation
 - ▶ Multi-vehicle tandem operation
- ▶ Reduced order modelling
 - ▶ Freeze and thawing of DoFs during run-time (e.g. in molecular dynamics)
- ▶ DoFs with high-gain control
 - ▶ Spacecraft actuator gimbals

Decomposition into active/passive DoFs and systems

- ▶ Refer to **prescribed** DoFs as **active**, and **non-prescribed** as **passive** DoFs
- ▶ Decompose the full system into active and passive systems
- ▶ Overall DoFs are the same as for the original system, but each has a sub-set of DoFs



Partitioned equations of motion

- ▶ With appropriate re-indexing the equations of motion can be partitioned based on the active and passive DoFs

$$\begin{pmatrix} \text{mass matrix of active system} & \text{Cross-coupling terms} \\ \boxed{M_{aa}} & \boxed{M_{ap}} \\ \text{Cross-coupling terms} & \text{Mass matrix of passive system} \\ \boxed{M_{ap}^T} & \boxed{M_{pp}} \end{pmatrix} \begin{bmatrix} \text{Known} \\ \boxed{\ddot{\theta}_a} \\ \boxed{\ddot{\theta}_p} \\ \text{unknown} \end{bmatrix} + \begin{bmatrix} \mathcal{C}_a \\ \mathcal{C}_p \end{bmatrix} = \begin{bmatrix} \text{unknown} \\ \boxed{\tau_a} \\ \boxed{\tau_p} \\ \text{known} \end{bmatrix}$$

- ▶ Have known and unknown quantities on both sides of the equations of motion.

$$\mathcal{M}_{ij} \triangleq H_i \phi M \phi^T H_j^T \quad \text{and} \quad \mathcal{C}_i \triangleq H_i \phi (\mathbf{b} + M \phi^T \mathbf{a})$$

Rearranged equations of motion

- Rearrange the partitioned EoM so that all unknowns are on and all knowns on the other

$$\underbrace{\begin{bmatrix} \tau_a \\ \ddot{\theta}_p \end{bmatrix}}_{\text{unknown}} = \begin{pmatrix} \mathcal{S}_{aa} & \mathcal{S}_{ap} \\ -\mathcal{S}_{ap}^T & \mathcal{S}_{pp} \end{pmatrix} \underbrace{\begin{bmatrix} \ddot{\theta}_a \\ \tau_p \end{bmatrix}}_{\text{known}} + \begin{bmatrix} \mathcal{C}_a - \mathcal{S}_{ap} \mathcal{C}_p \\ -\mathcal{S}_{pp} \mathcal{C}_p \end{bmatrix}$$

- The mass matrix inverse of the passive system shows up in each of the $\mathcal{S}$ variables

$$\mathcal{S}_{aa} \triangleq \mathcal{M}_{aa} - \mathcal{M}_{ap} \mathcal{M}_{pp}^{-1} \mathcal{M}_{ap}^T$$

$$\mathcal{S}_{ap} \triangleq \mathcal{M}_{ap} \mathcal{M}_{pp}^{-1}$$

$$\mathcal{S}_{pp} \triangleq \mathcal{M}_{pp}^{-1}$$

ATBI recursion for passive system

- The ATBI quantities for the passive system can be computed as usual by a gather recursion

$$P^+(0) = 0$$

for $k = 1 \dots n$

$$P(k) = \phi(k, k-1) P^+(k-1) \phi^T(k, k-1) + M(k)$$

if $k \notin I_p$

Skips the active dof hinge

$$P^+(k) = P(k)$$

else

Normal ATBI steps for passive DoF

$$\mathcal{D}_p(k) = H_p(k) P(k) H_p^T(k)$$

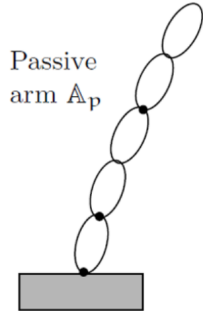
$$\mathcal{G}_p(k) = P(k) H_p^T(k) \mathcal{D}_p^{-1}(k)$$

$$\mathcal{K}_p \bar{\tau}_p(k) = I - \mathcal{G}_p(k) H_p(k)$$

$$P^+(k) = \bar{\tau}_p(k) P(k)$$

end if

end loop



Passive system ATBI spatial operators

- ▶ The spatial operators are the usual ATBI spatial operators except they are restricted to the passive DoFs.

$$\mathcal{D}_p \triangleq H_p P H_p^T \quad \in \mathbb{R}^{\mathcal{N}_p \times \mathcal{N}_p}$$

$$\mathcal{G}_p \triangleq P H_p^T \mathcal{D}_p^{-1} \quad \in \mathbb{R}^{6n \times \mathcal{N}_p}$$

$$\mathcal{K}_p \triangleq \epsilon_\phi \mathcal{G} \quad \in \mathbb{R}^{6n \times \mathcal{N}_p}$$

$$\bar{\tau}_p \triangleq I - \mathcal{G}_p H_p \quad \in \mathbb{R}^{6n \times 6n}$$

$$\epsilon_\psi \triangleq \epsilon_\phi \bar{\tau}_p \quad \in \mathbb{R}^{6n \times 6n}$$

- ▶ Riccati equation for the passive system

$$M = P - \epsilon_\psi^T P \epsilon_\psi = P - \epsilon_\phi P \epsilon_\psi^T$$

Passive system mass matrix inversion

- ▶ Since $\mathcal{M}_{pp}$ is a mass matrix it can be factorized and inverted as previously
- ▶ Newton Euler factorization of the mass matrix

$$\mathcal{M}_{pp} = H_p \phi M \phi^T H_p^T$$

- ▶ Innovations factorization of the mass matrix

$$\mathcal{M}_{pp} = (I + H_p \phi \mathcal{K}_p) \mathcal{D}_p (I + H_p \phi \mathcal{K}_p)^T$$

- ▶ Mass matrix inverse

$$(I + H_p \phi \mathcal{K}_p)^{-1} = (I - H_p \psi \mathcal{K}_p)$$
$$\mathcal{M}_{pp}^{-1} = (I - H_p \psi \mathcal{K}_p)^T \mathcal{D}_p^{-1} (I - H_p \psi \mathcal{K}_p)$$

Expression simplifications

$$\begin{bmatrix} \tau_a \\ \ddot{\theta}_p \end{bmatrix} = \begin{pmatrix} \mathcal{S}_{aa} & \mathcal{S}_{ap} \\ -\mathcal{S}_{ap}^T & \mathcal{S}_{pp} \end{pmatrix} \begin{bmatrix} \ddot{\theta}_a \\ \tau_p \end{bmatrix} + \begin{bmatrix} \mathcal{C}_a - \mathcal{S}_{ap} \mathcal{C}_p \\ -\mathcal{S}_{pp} \mathcal{C}_p \end{bmatrix}$$

► Claim

$$\mathcal{S}_{pp} = (I - H_p \psi \mathcal{K}_p)^T \mathcal{D}^{-1} (I - H_p \psi \mathcal{K}_p)$$

$$\begin{aligned} \mathcal{S}_{ap} &= H_a (\psi \mathcal{K}_p + P \psi^T H_p^T \mathcal{D}^{-1} (I - H_p \psi \mathcal{K}_p)) \\ &= H_a ((\psi - P \Omega) \mathcal{K}_p + P \psi^T H_p^T \mathcal{D}_p^{-1}) \end{aligned}$$

$$\begin{aligned} \mathcal{S}_{aa} &= H_a (\psi M \psi^T - P \Omega P) H_a^T \\ &= H_a ((\psi - P \Omega) P + P \tilde{\psi}^T) H_a^T \end{aligned}$$

$$\Omega \triangleq \psi^T H_p^T \mathcal{D}_p^{-1} H_p \psi$$

operational space inertia (more later)



Derivation of $\mathcal{S}_{ap}$ expression

$$\begin{aligned}\mathcal{S}_{ap} &= \mathcal{M}_{ap} \mathcal{M}_{pp}^{-1} \\ &= H_a \phi M \phi H_p^T (I - H_p \psi \mathcal{K}_p)^T \mathcal{D}^{-1} (I - H_p \psi \mathcal{K}_p) \\ &= H_a \phi M \psi^T H_p^T \mathcal{D}^{-1} (I - H_p \psi \mathcal{K}_p) \\ &= H_a (\phi \mathcal{K}_p + P \psi^T H_p^T \mathcal{D}^{-1}) (I - H_p \psi \mathcal{K}_p) \\ &= H_a (\psi \mathcal{K}_p + P \psi^T H_p^T \mathcal{D}^{-1} (I - H_p \psi \mathcal{K}_p)) \\ &= H_a ((\psi - P \Omega) \mathcal{K}_p + P \psi^T H_p^T \mathcal{D}^{-1})\end{aligned}$$

Derivation of $\mathcal{S}_{aa}$ expression

$$\begin{aligned}\mathcal{S}_{aa} &= \mathcal{M}_{aa} - \mathcal{M}_{ap} \mathcal{M}_{pp}^{-1} \mathcal{M}_{ap}^T \\ &= \mathcal{M}_{aa} - \mathcal{S}_{ap} \mathcal{M}_{ap}^T \\ &= H_a \left(\phi - \psi \mathcal{K}_p H_p \phi - P \psi^T H^T \mathcal{D}_p^{-1} (I - H_p \psi \mathcal{K}_p) H_p \phi \right) M \phi^T H_a^T \\ &= H_a \left(\phi - (\phi - \psi) - P \psi^T H^T \mathcal{D}_p^{-1} H_p \psi \right) M \phi H_a^T \\ &= H_a \left(\psi - P \psi^T H^T \mathcal{D}_p^{-1} H_p \psi \right) M \phi^T H_a^T \\ &= H_a (\psi \cdot P \Omega) M \phi^T H_a^T \\ &= H_a \left((\psi - P \Omega) P + P \tilde{\psi}^T \right) H_a^T\end{aligned}$$

Hybrid Dynamics Solution



Operator expressions for solution

- ▶ Defining operators similar to the $O(N)$ recursion operators

$$\underline{\mathbf{a}} \triangleq H_a^T \ddot{\theta}_a + \mathbf{a}$$

$$\epsilon_p \triangleq \tau_p - H_p \xi$$

$$\xi \triangleq \psi (\mathcal{K}_p \tau_p + \mathbf{b} + P \underline{\mathbf{a}})$$

$$\nu_p \triangleq \mathcal{D}_p^{-1} \epsilon_p$$

- ▶ claim:

$$\alpha = \psi^T (H_p^T \nu_p + \underline{\mathbf{a}})$$

$$\ddot{\theta}_p = (I - H_p \psi \mathcal{K}_p)^T \nu_p - \mathcal{K}_p^T \psi^T \underline{\mathbf{a}} = \nu_p - \mathcal{K}_p^T \alpha$$

$$\tau_a = H_a \left(\xi + P \left(\psi^T H_p^T \nu_p + \tilde{\psi}^T \underline{\mathbf{a}} \right) \right) = H_a (P (\alpha - \underline{\mathbf{a}}) + \xi)$$

Derivation of $\ddot{\theta}_p$ expression

$$\begin{bmatrix} \tau_a \\ \ddot{\theta}_p \end{bmatrix} = \begin{pmatrix} \mathcal{S}_{aa} & \mathcal{S}_{ap} \\ -\mathcal{S}_{ap}^T & \mathcal{S}_{pp} \end{pmatrix} \begin{bmatrix} \ddot{\theta}_a \\ \tau_p \end{bmatrix} + \begin{bmatrix} \mathcal{C}_a - \mathcal{S}_{ap} \mathcal{C}_p \\ -\mathcal{S}_{pp} \mathcal{C}_p \end{bmatrix} \Rightarrow \ddot{\theta}_p = \mathcal{S}_{pp} (\tau_p - \mathcal{C}_p) - \mathcal{S}_{ap}^T \ddot{\theta}_a$$

► Thus

$$\begin{aligned} \ddot{\theta}_p &= \mathcal{S}_{pp} (\tau_p - \mathcal{C}_p) - \mathcal{S}_{ap}^T \ddot{\theta}_a \\ &= (I - H_p \psi \mathcal{K}_p)^T \mathcal{D}_p^{-1} (\tau_p - H_p \psi (\mathcal{K}_p \tau_p + \mathfrak{b} + M \phi^T \mathfrak{a})) \\ &\quad - \left(\mathcal{K}_p^T \psi^T + (I - H_p \psi \mathcal{K}_p)^T \mathcal{D}_p^{-1} H_p \psi P \right) H_a^T \ddot{\theta}_a \\ &= (I - H_p \psi \mathcal{K}_p)^T \mathcal{D}_p^{-1} (\tau_p - H_p \psi (\mathcal{K}_p \tau_p + P \mathfrak{a} + \mathfrak{b})) - \mathcal{K}_p^T \psi^T \mathfrak{a} \\ &\quad - \left(\mathcal{K}_p^T \psi^T + (I - H_p \psi \mathcal{K}_p)^T \mathcal{D}_p^{-1} H_p \psi P \right) H_a^T \ddot{\theta}_a \end{aligned}$$

Derivation of $\ddot{\theta}_p$ expression (contd)

$$\begin{bmatrix} \tau_a \\ \ddot{\theta}_p \end{bmatrix} = \begin{pmatrix} \mathcal{S}_{aa} & \mathcal{S}_{ap} \\ -\mathcal{S}_{ap}^T & \mathcal{S}_{pp} \end{pmatrix} \begin{bmatrix} \ddot{\theta}_a \\ \tau_p \end{bmatrix} + \begin{bmatrix} \mathcal{C}_a - \mathcal{S}_{ap} \mathcal{C}_p \\ -\mathcal{S}_{pp} \mathcal{C}_p \end{bmatrix} \Rightarrow \ddot{\theta}_p = \mathcal{S}_{pp} (\tau_p - \mathcal{C}_p) - \mathcal{S}_{ap}^T \ddot{\theta}_a$$

► Thus

$$\begin{aligned} \ddot{\theta}_p &= (I - H_p \psi \mathcal{K}_p)^T \mathcal{D}_p^{-1} (\tau_p - H_p \psi (\mathcal{K}_p \tau_p + P \underline{\mathbf{a}} + \underline{\mathbf{b}})) - \mathcal{K}_p^T \psi^T \underline{\mathbf{a}} \\ &= (I - H_p \psi \mathcal{K}_p) \nu_p - \mathcal{K}_p^T \psi^T \underline{\mathbf{a}} \\ &= \nu_p - \mathcal{K}_p^T \alpha \end{aligned}$$

Derivation of body accelerations expression

$$\begin{aligned}\alpha &= \phi^T \left(H^T \ddot{\theta} + \underline{\mathbf{a}} \right) \\ &= \phi^T \left(H_p^T \ddot{\theta}_p + \underline{\mathbf{a}} \right) \\ &= \phi^T H_p^T \left((I - H_p \psi \mathcal{K}_p)^T \nu_p - \mathcal{K}_p^T \psi^T \underline{\mathbf{a}} \right) + \phi^T \underline{\mathbf{a}} \\ &= \psi^T \left(H_p^T \nu_p + \underline{\mathbf{a}} \right)\end{aligned}$$

Derivation of $\mathcal{S}_{ap} \mathcal{C}_p$ expression

$$\begin{aligned}\mathcal{S}_{ap} \mathcal{C}_p &= H_a \left(\psi \mathcal{K}_p + P \psi^T H_p^T \mathcal{D}^{-1} (I - H_p \psi \mathcal{K}_p) \right) H_p \phi (\mathfrak{b} + M \phi^T \mathfrak{a}) \\&= H_a \left((\phi - \psi) + P \psi^T H_p \mathcal{D}_p^{-1} H_p \psi \right) (\mathfrak{b} + M \phi^T \mathfrak{a}) \\&= \mathcal{C}_a - H_a (\psi - P \Omega) (M \phi^T \mathfrak{a} + \mathfrak{b}) \\&= \mathcal{C}_a - H_a (\psi - P \Omega) M \phi^T \mathfrak{a} - H_a (\psi - P \omega) \mathfrak{b} \\&= \mathcal{C}_a - H_a \left(\tilde{\psi} P + P \phi^T - P (\phi^T - \psi^T + \Omega P) \right) \mathfrak{a} - H_a (\psi - P \Omega) \mathfrak{b} \\&= \mathcal{C}_a H_a (\psi - P \Omega) (P \mathfrak{a} + \mathfrak{b}) - H_a P \tilde{\psi}^T \mathfrak{a}\end{aligned}$$

Derivation of τ_a expression

$$\tau_a = \mathcal{S}_{aa} \ddot{\theta}_a + \mathcal{S}_{ap} (\tau_p - \mathcal{C}_p) + \mathcal{C}_a$$

$$\begin{aligned}\tau_a &= \mathcal{S}_{aa} \ddot{\theta}_a + \mathcal{S}_{ap} (\tau_p - \mathcal{C}_p) + \mathcal{C}_a \\ &= H_a \left((\psi - P \Omega) P + P \tilde{\psi}^T \right) H_a^T \ddot{\theta}_a + H_a \left((\psi - P \Omega) \mathcal{K}_p + P \psi^T H_a^T \mathcal{D}^{-1} \right) \tau_p \\ &\quad + H_a (\psi + P \Omega) (P \mathfrak{a} + \mathfrak{b}) + H_a P \tilde{\psi}^T \mathfrak{a} \\ &= H_a \left((\psi - P \Omega) \left(P H_a^T \ddot{\theta}_a + \mathcal{K}_p \tau_p + (P \mathfrak{a} + \mathfrak{b}) \right) \right. \\ &\quad \left. + P \left(\tilde{\psi}^T H_a^T \ddot{\theta}_a + \psi^T H_a^T \mathcal{D}_p^{-1} \tau_p + \tilde{\psi}^T \mathfrak{a} \right) \right)\end{aligned}$$

Derivation of τ_a expression (contd)

$$\tau_a = \mathcal{S}_{aa} \ddot{\theta}_a + \mathcal{S}_{ap} (\tau_p - \mathcal{C}_p) + \mathcal{C}_a$$

$$\begin{aligned}\tau_a &= H_a \left((I - P \psi^T H_p^T \mathcal{D}^{-1} H_p) \xi + P \left(\tilde{\psi}^T \underline{\mathbf{a}} + \psi^T H_p^T \mathcal{D}_p^{-1} \tau_p \right) \right) \\ &= H_a \left(\xi + P \psi^T H_p^T \mathcal{D}^{-1} (\tau_p - H_p \xi) + P \tilde{\psi}^T \underline{\mathbf{a}} \right) \\ &= H_a \left(\xi + P \psi^T H_p^T \mathcal{D}^{-1} \epsilon + P \tilde{\psi}^T \underline{\mathbf{a}} \right) \\ &= H_a \left(\xi + P \left(\psi^T H_p^T \nu + \tilde{\psi}^T \underline{\mathbf{a}} \right) \right) \\ &= H_a (\xi + P (\alpha - \underline{\mathbf{a}}))\end{aligned}$$

Hybrid Dynamics Algorithm



Operator expressions for solution

- These expressions are very similar to regular ATBI forward dynamics and also map into similar $O(N)$ ATBI gather/scatter recursions

Active gen accels term

$$\underline{\mathbf{a}} \triangleq \boxed{H_a^T \ddot{\theta}_a} + \mathbf{a}$$

$$\xi \triangleq \psi (\mathcal{K}_p \tau_p + \mathbf{b} + P \underline{\mathbf{a}})$$

$$\alpha = \psi^T (H_p^T \nu_p + \underline{\mathbf{a}})$$

$$\ddot{\theta}_p = \nu_p - \mathcal{K}_p^T \alpha$$

$$\tau_a = H_a (P (\alpha - \underline{\mathbf{a}}) + \xi)$$

$$\epsilon_p \triangleq \tau_p - H_p \xi$$

$$\nu_p \triangleq \mathcal{D}_p^{-1} \epsilon_p$$

$O(N)$ Hybrid dynamics algorithm

► Virtually the same structure as the regular $O(N)$ ATBI algorithm

```

 $\xi(0) = 0$ 
for  $k = 1 \dots n$ 
  if  $k \in I_a$ 
    Altered steps for the active/prescribed DoF
     $\xi(k) = \phi(k, k-1) \xi^+(k-1) + (k) + P(k) \left( H^T(k) \ddot{\theta}_a(k) + \mathbf{a}(k) \right)$ 
     $\xi^+(k) = \xi(k)$ 
  else
     $\xi(k) = \phi(k, k-1) \xi^+(k-1) + (k) + P(k) \mathbf{a}(k)$ 
     $\epsilon_p(k) = \tau_p(k) - H(k) \xi(k)$ 
     $\xi^+(k) = \xi(x) + \mathcal{G}_p(k) \epsilon_p(k)$ 
     $\nu_p(k) = \mathcal{D}_p^{-1} \epsilon_p(k)$ 
  end if
end loop
```

$O(N)$ Hybrid dynamics algorithm

- Virtually the same structure as the regular $O(N)$ ATBI algorithm

$$\alpha^+(n+1) = 0$$

for $k = n \dots 1$

$$\alpha^+(k) = \phi^T(k+1, k) \alpha(k+1)$$

if $k \in I_a$

Altered steps for the active/prescribed DoF

$$\mathfrak{f}(k) = P(k) \alpha^+(k) + \xi(k)$$

$$\tau_a(k) = H(k) \mathfrak{f}(k)$$

else

$$\ddot{\theta}(k) = \nu_p(k) - \mathcal{G}_p^T(k) \alpha^+(k)$$

end if

$$\alpha(k) = \alpha^+(k) + H^T(k) \ddot{\theta}(k) + \mathfrak{a}(k)$$

end loop

Recall: CRB-based Inverse dynamics algorithm

$$\mathbf{f} = \mathcal{R} \alpha + \mathbf{y} \quad \mathbf{y} \triangleq \phi \left(\mathbf{b} + \epsilon_{\phi} \mathcal{R} \left(H^T \ddot{\theta} + \mathbf{a} \right) \right)$$

- ▶ Use CRB gather algorithm to compute the CRB spatial inertias
- ▶ Compute the $\mathbf{y}$ values via a gather algorithm

$$\begin{aligned} \mathbf{y}^+(0) &= \mathbf{0} \\ \text{for } k &= 1 \dots n \\ \mathbf{y}(k) &= \phi(k, k-1) \mathbf{y}^+(k-1) + \mathbf{b}(k) \\ \mathbf{y}^+(k) &= \mathbf{y}(k) + \mathcal{R}(k) \left(H^T(k) \ddot{\theta}(k) + \mathbf{a}(k) \right) \\ \text{end loop} \end{aligned}$$

- ▶ The hybrid dynamics steps for the active dofs switch to this CRB inverse dynamics steps for active dofs!
- ▶ Compute the generalized forces

$$\tau(k) = H(k) \mathbf{f}(k) = H(k) (\mathcal{R}(k) \alpha(k) + \mathbf{y}(k))$$

Comments on $O(N)$ hybrid dynamics algorithms

- ▶ The hybrid dynamics algorithm is very similar in structure to the gather/scatter sweeps of the ATBI forward dynamics algorithm
- ▶ The computational cost scales linearly with the number of DoFs
- ▶ The algorithm elegantly uses ATBI forward dynamics algorithm steps for passive DoFs and the CRB based inverse dynamics algorithm steps for the active DoFs
- ▶ When all DoFs are passive we get the usual ATBI forward dynamics
- ▶ When all DoFs are active we get the CRB inverse dynamics
- ▶ Can change the mode of a hinge at any time from active to passive (at run-time) and the algorithm automatically accommodates this

Examples where run-time change for prescribed motion might be used

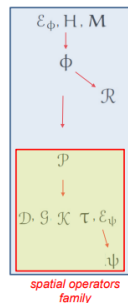
- ▶ Consider the separation or docking scenario between pair of vehicles
- ▶ Model the system as one system with a 6DoF connection hinge
- ▶ When docked, set the hinge to prescribed motion with zero prescribed accelerations – essentially locks the hinge
- ▶ For undocking, disable the prescribed motion setting to allow the vehicles to move apart.

How far have we come?



Spatial operators (no new operators)

- ▶ Velocity expression
 - ▶ Jacobian
 - ▶ Mass matrix Newton Euler factorization
 - ▶ Lyapunov equation for CRBs
 - ▶ Mass matrix decomposition
 - ▶ Riccati equation for ATBI
 - ▶ Several operator identities
 - ▶ Mass matrix Innovations factorization
 - ▶ Mass matrix determinant
 - ▶ Mass matrix inverse and factorization
- ▶ Have started to build up a vocabulary of spatial operators that can be used to express and manipulate the structure of dynamics quantities
- ▶ Now we can see algebra part of SOA from the analytical transformations and simplifications possible using the operators.



Recursive Computational Algorithms

- ▶ Low-cost scatter/gather algorithms for examination of the spatial operator expressions has been developed
 - ▶ $O(N)$ Gather and scatter recursions pattern
 - ▶ $O(N)$ Body velocities scatter recursion
 - ▶ $O(N)$ CRBs gather recursion
 - ▶ $O(N^2)$ mass matrix computation
 - ▶ $O(N)$ Newton Euler scatter/gather inverse dynamics
 - ▶ $O(N^2)$ inverse dynamics based mass matrix
 - ▶ $O(N)$ CRBs based inverse dynamics
 - ▶ $O(N)$ ATBI gather recursion
 - ▶ $O(N^2)$ forward dynamics
 - ▶ $O(N)$ ATBI forward dynamics
 - ▶ $O(N)$ hybrid dynamics

Summary

- ▶ Used the operator expression for the mass matrix inverse to develop the $O(N)$ ATBI forward dynamics algorithm
- ▶ Described simple way to obtain inter-body forces if desired
- ▶ Developed extensions for handling gravity and external forces
- ▶ Developed $O(N)$ generalized hybrid dynamics algorithm
 - ▶ Elegant combination of ATBI forward dynamics and CRB inverse dynamics